Today's goals
Complete any 3 learning items · 3/3
Complete 3 readings · 3/3
Watch a video

Module 1
Foundations of Multimodal AI

Welcome to the Course
Video: Course Introduction
Video • 4 min
Reading: Course Overview
Reading • 3 min
RAG and Agentic AI Professional Certificate Overview
Video • 5 min
Helpful Tips for Course Completion
Ungraded Plugin • 3 min

Introduction to Multimodal AI: Text and Speech Processing
Introduction to Multimodal AI
Video • 8 min
Reading: What is Multimodal Generative AI and Why Does It Matter?
Ungraded Plugin • 5 min
Reading: What is Computer Vision?
Ungraded Plugin • 7 min
Text-to-Speech Technologies
Video • 8 min
Speech-to-Text Technologies
Video • 7 min
Reading: Text Processing, Speech Processing, and Text-to-Speech
Ungraded Plugin • 7 min
Lab: Use Mistral and gTTS to Create Your Personal Storyteller
Ungraded App Item • 30 min
Reading: Challenges in Multimodal AI Integration
Ungraded Plugin • 5 min
Lab: Build a Meeting Assistant with Whisper, LangChain, & Gradio
Ungraded App Item • 45 min
Practice Quiz: Introduction to Multimodal AI: Text and Speech Processing
Practice Assignment • 15 min

Module Summary and Evaluation
Reading: Summary and Highlights
Reading • 2 min
Cheat Sheet: Foundations of Multimodal AI
Ungraded Plugin • 5 min
Graded Quiz: Foundations of Multimodal AI
Graded Assignment • 21 min

Module 2
Integrating Visual and Video Modalities

Module 3
Advanced Multimodal Applications

Reading: Course Overview

0:14 / 3:42 1x

Welcome to the **Build Multimodal Generative AI Applications** course!

This intermediate-level course is designed for AI developers, data scientists, and engineers looking to expand their expertise in building applications that leverage multimodal generative AI. You'll gain hands-on experience integrating text, speech, images, and video to develop practical, cross-modal applications using powerful models and frameworks.

Throughout this course, you'll explore how to build applications that transform and generate across modalities—such as turning speech into text, text into images, or images into captions—using models like Meta's Llama 4, OpenAI Whisper, and GPT Image. You'll work with tools such as Gradio, Flask, LangChain, and Hugging Face to prototype and deploy user-facing multimodal AI systems.

You'll begin by mastering the fundamentals of multimodal AI, gaining experience with speech-to-text and text-to-speech applications. Then, you'll learn how to generate and caption images and videos using advanced generative models. Finally, you'll build real-world multimodal applications—such as chatbots, personal assistants, and recommendation systems—using cross-modal retrieval and generation techniques.

This course is part of the **IBM RAG and Agentic AI Professional Certificate**, aimed at equipping learners with cutting-edge skills to develop next-generation AI systems that understand and generate multiple forms of media.

Prerequisites

To succeed in this course, you should have:

- Proficiency in Python
- Prior exposure to Flask, Gradio, and LangChain
- A working knowledge of generative AI fundamentals and frameworks like Hugging Face

Learning objectives

After completing this course, you will be able to:

- Explain key challenges and integration techniques in multimodal AI
- Build applications using models like Whisper, GPT Image, and Llama 4
- Generate and caption media across text, image, speech, and video modalities
- Implement chatbots, recommendation engines, and other applications that use multimodal retrieval
- Deploy AI applications using Flask and Gradio

Course Modules

Module 1: Foundations of Multimodal AI

- Introduction to Multimodal AI
- Text-to-Speech Technologies
- Speech-to-Text Technologies
- Build applications based on AI-generated audiobooks and automated meeting assistants using tools such as OpenAI Whisper, Mistral, gTTS, Gradio, and LangChain

Module 2: Integrating Visual and Video Modalities

- Understanding Image Captioning with Meta's Llama
- Text-to-Video Generation with OpenAI's Sora
- Introduction to image-to-text and text-to-video capabilities
- Hands-on labs with image generation using GPT Image and image captioning using IBM watsonx and Granite

Module 3: Advanced Multimodal Applications

- Introduction to multimodal retrieval-augmented generation (MM-RAG)
- Multimodal Chatbots and QA Systems
- Build AI apps for fashion matching, visual search, and diet recommendation
- Use Flask and Gradio to deploy interactive multimodal applications

Tools and Frameworks

You'll gain experience with:

- **GPT Image, Whisper**, and **Llama 4** for multimodal generation
- **Gradio** and **Flask** for building user interfaces
- **LangChain** for chaining multimodal processing tasks
- **Hugging Face** for deploying and accessing pre-trained models

Tips for Success

- Stay hands-on: Engage deeply with labs and projects
- Think creatively: Use multimodal AI in real-world applications
- Leverage communities and documentation for open-source tools

Ready to explore how AI sees, hears, speaks, and creates across modalities? Let's begin your journey into **multimodal generative AI**.

Mark as completed

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